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RESEARCH INTERESTS

Model Quantization for Training and Inference, Data-free Neural Network Compression, Generative Models

EDUCATION

- **Seoul National University** Seoul, Korea
Ph.D. in Electrical and Computer Engineering, GPA 4.15/4.3 September 2023 - February 2026
- **Yonsei University** Seoul, Korea
M.S. in Computer Science, GPA 4.20/4.3 September 2020 - August 2023
- **Yonsei University** Seoul, Korea
B.S. in Computer Science, GPA 3.81/4.3 March 2016 - August 2020
- **American University** Washington D.C., USA
Study Abroad Program, Computer Science January 2019 - May 2019

PUBLICATIONS

- (ECCV 2026) STARLINC: Satellite Trail Artifact Removal using Inter-Frame Correlation
Shingeon Kim, Hyeyoon Lee, Dain Kwon, **Kanghyun Choi**, Sunjong Park, Mi-Ryang Kim, Jeong-Eun Lee, Jinho Lee
- (DAC 2026) TopGQ: Fast GNN Post-Training Quantization Leveraging Topology Information
Dain Kwon, **Kanghyun Choi**, Hyeyoon Lee, Sunjong Park, Seoyong Lee, Sukjin Kim, Jinho Lee
- (DAC 2026) SePARate: Segmenting Patterns from Defects in Wafer Manufacturing Using Weak Supervision
Dain Kwon, Changmin Shin, Sunjong Park, **Kanghyun Choi**, Hyeyoon Lee, Jaewon Jang, Minseok Choi, Jinho Lee
- (NeurIPS 2025) FALQON: Accelerating LoRA Fine-tuning with Low-bit Floating-point Arithmetic
Kanghyun Choi, Hyeyoon Lee, SunJong Park, Dain Kwon, and Jinho Lee
- (IEEE TCAD) DANCE++: Differentiable Accelerator/Network Co-Exploration with Hard Constraints and Data-Free Training for Real-World Scenarios
Kanghyun Choi, Deokki Hong, Hyeyoon Lee, Joonsang Yu, Noseong Park, Youngsok Kim, and Jinho Lee
- (AAAI 2025) MimiQ: Low-Bit Data-Free Quantization of Vision Transformers with Encouraging Inter-Head Attention Similarity
Kanghyun Choi, Hyeyoon Lee, Dain Kwon, SunJong Park, Kyuyeun Kim, Noseong Park, Jonghyun Choi, and Jinho Lee
- (ICML 2024) DataFreeShield: Defending Adversarial Attacks without Training Data
Hyeyoon Lee, **Kanghyun Choi**, Dain Kwon, SunJong Park, Mayoore Jaiswal, Noseong Park, Jonghyun Choi, and Jinho Lee
- (DAC 2024) Fast Adversarial Training with Dynamic Batch-level Attack Control
Jaewon Jung, Jaeyong Song, Hongsun Jang, Hyeyoon Lee, **Kanghyun Choi**, Noseong Park, Jinho Lee
- (CVPR 2022 Oral) It's All In the Teacher: Zero-Shot Quantization Brought Closer to the Teacher
Kanghyun Choi, Hyeyoon Lee, Deokki Hong, Joonsang Yu, Noseong Park, Youngsok Kim, Jinho Lee
- (DAC 2022) Enabling Hard Constraints in Differentiable Neural Network and Accelerator Co-Exploration
Deokki Hong, **Kanghyun Choi**, Hyeyoon Lee, Joonsang Yu, Noseong Park, Youngsok Kim, Jinho Lee
- (NeurIPS 2021) Qimera: Data-free Quantization with Synthetic Boundary Supporting Samples
Kanghyun Choi, Deokki Hong, Noseong Park, Youngsok Kim, Jinho Lee
- (DAC 2021) DANCE: Differentiable Accelerator/Network Co-Exploration
Kanghyun Choi¹, Deokki Hong¹, Hojae Yoon¹, Joonsang Yu, Youngsok Kim, Jinho Lee

AWARDS

- **The 32nd Samsung Humantech Paper Award:** February 2026 *Gold Prize, Signal Processing*
- **AI Model Benchmarking Competition:** October 2025 *3rd Prize, IEEE/ACM MICRO 2025 AI-BMT Workshop*
- **The 28th Samsung Humantech Paper Award:** February 2022 *Silver Prize, Computer Science and Engineering*
- **High Honors at Graduation (Top 3% of class):** August 2020

TEACHING EXPERIENCE

- **Programming Methodology (430.211)**: Head Teaching Assistant, Spring 2024
- **Digital System Design and Practice (430.315A)**: Teaching Assistant, Fall 2023
- **Multi-core and GPU Programming (CSI4119)**: Teaching Assistant, Spring 2021, 2022
- **Logic Circuit Design (CSI2111)**: Teaching Assistant, Fall 2020

PROJECTS

- **Large Language Model Training with FP8 Arithmetic**
Model Optimization Program, Google 2024-2025
- **Data-Free Quantization Framework for Vision Transformers**
Model Optimization Program, Google 2023-2024
- **Accelerating Diffusion Models for Landscape Generation**
Electronics and Telecommunications Research Institute (ETRI) 2023-2024
- **Semantic Modification Method for High-resolution Face Images**
Electronics and Telecommunications Research Institute (ETRI) 2022
- **High-resolution Face Image Generation by Transformer-based GAN**
Electronics and Telecommunications Research Institute (ETRI) 2021
- **Fast Distributed Deep Neural Network Training**
Korea Institute of Industrial Technology (KITECH) 2020

ACADEMIC SERVICES

- **Reviewer**: ICLR, ICML, NeurIPS, CVPR, ICCV, ECCV, ACCV, AAAI, TMLR, IEEE SPL

SKILLS

- **Python, C, C++, $\LaTeX$**
- **Pytorch, Pandas, SciPy**
- **Korean** (Native), **English** (Fluent), **Japanese** (Intermediate)